Harsha Vardhan Yellela

United States | +1-248-497-9965 | harsha.yellela@gmail.com | har5ha.in | LinkedIn | GitHub

SUMMARY

Software Engineer with hands-on experience building **scalable backend systems**, **RESTful APIs**, and **cloud-native microservices**. Proficient in **Python (FastAPI)**, **Go (Gin)**, and **AWS services (Lambda, ECS, DynamoDB)**. Built production systems handling **94+ serverless functions**, **third-party integrations (Stripe, Twilio, DocuSign)**, and **Kubernetes deployments**. Strong foundation in **API design**, **containerization**, **CI/CD pipelines**, and **database optimization**.

EXPERIENCE

Student Consultant – Software Engineering **Feb 2026 – Present**
Open Avenues (Build Fellowship) *Remote*

- Building an end-to-end language model system in Python/PyTorch — custom **web scraping pipeline**, data preprocessing (~3.2M chars), BPE tokenizer training, multi-GPU model training, and production deployment on **HuggingFace Spaces**.
- Implemented a **CLI tool** (gpt.py) with `--train`, `--eval`, `--interactive`, and `--custom-bpe` modes; engineered clean interfaces for training, inference, checkpoint save/load, and tokenizer switching.
- Applying software engineering principles — modular design, reproducible experiment tracking, train/val/test splits — to build a GPT transformer **entirely from scratch** without high-level ML frameworks.

Graduate Research Assistant – Agentic AI **Jan 2025 – Dec 2025**
Lawrence Technological University *Southfield, MI*

- Deployed persistent **backend services** on **AWS Fargate** and **Amazon EKS**, integrating **OpenSearch Serverless** for semantic search and document retrieval.
- Built **multi-agent systems** using **Python (CrewAI + LangChain)** with **RESTful APIs** for workflow automation and intelligent decision-making.
- Designed **hybrid service pipelines** achieving **70% reduction in manual process time** through automation and API orchestration.

Infor India Pvt. Ltd. **Apr 2022 – Dec 2023**
LN Technical Consultant *Hyderabad, India*

- Developed **modular, production-ready tools** for global clients (**Ferrari, Boeing, Triumph**) by extending **Infor LN ERP** workflows for inventory and manufacturing.
- Integrated **AWS S3, Lambda, and API Gateway** with enterprise systems to enable **asynchronous data pipelines**, reducing **batch processing delays by ~40%**.
- **Containerized** business logic services using **Docker** and orchestrated deployments, improving **consistency** across client environments.

Junior Java Trainee (PEP Program) **Sep 2020 – Jul 2021**
EPAM Systems *Hyderabad, India*

- Completed intensive **Java Core** training covering OOP design, Collections, Multithreading, and Streams API; applied **REST API design patterns** and **JUnit-style unit testing** across all practice modules.
- Built backend modules using **Spring Boot fundamentals**; practiced **Git-based version control** and operated in structured **Agile/Scrum** sprints with peer code reviews.

TECHNICAL SKILLS

- **Languages:** Python, Go, TypeScript, JavaScript, SQL, Bash | **Core:** Data Structures, Algorithms, OOP
- **Backend Frameworks:** FastAPI, Flask, Gin (Go), Express.js | Pydantic, JWT, CORS, HATEOAS
- **Cloud (AWS):** Lambda, ECS Fargate, EC2, API Gateway, S3, DynamoDB, ECR, SQS, SageMaker
- **Databases:** PostgreSQL, MySQL, SQLite, DynamoDB, MongoDB | **Caching:** Redis | **Vector:** Qdrant, pgvector
- **DevOps:** Git, Linux, Docker, Kubernetes (EKS), Terraform, Jenkins, GitHub Actions, CI/CD Pipelines
- **Practices:** Agile/Scrum, Unit Testing, Integration Testing, Code Reviews, RESTful API Design
- **Integrations:** Stripe, Twilio, DocuSign, QuickBooks, OpenAPI/Swagger

PROJECTS

AI Resume Builder (Resumade.in) – Serverless Backend **Dec 2025**
FastAPI, AWS Lambda, DynamoDB, GPT-4, Gemini, BeautifulSoup

- Architected **serverless backend** using **FastAPI + AWS Lambda + DynamoDB** with **95%+ uptime** and **sub-2s PDF/Word generation**.
- Integrated **multi-LLM pipelines (GPT-4, Gemini, DeepSeek)** with custom prompts; implemented **LinkedIn job scraping** for tailored resume optimization.
- Configured **JWT + Google OAuth authentication** and **CI/CD via GitHub Actions**, reducing manual resume effort by **80%**.

Blindspot – Cognitive Bias Detector Chrome Extension **Jan 2026**
JavaScript, Chrome Extension (Manifest V3), Claude API, Service Workers | GitHub

- Built **Chrome Extension** with **Manifest V3 architecture** using **Service Workers**, **Content Scripts**, and **Context Menu API** for real-time webpage analysis.
- Integrated **Claude API** for AI-powered cognitive bias detection; implemented **keyboard shortcuts (Ctrl+Shift+B)** and **right-click context menu** triggers.
- Designed **proactive intervention system** monitoring high-risk sites (e-commerce, subscriptions) with **content script injection** for universal webpage support.

Online Learning Portal – Microservices CI/CD Platform **Nov 2025**
FastAPI, Jenkins, Docker, Kubernetes, AWS ECS Fargate | GitHub

- Built **6-microservice platform** (user, course, enrollment, payment, notification, API gateway) using **FastAPI** with **Docker containerization**.
- Implemented **branch-per-service CI strategy** with **Jenkins Multibranch Pipeline**; automated Docker builds with **commit hash tagging**.
- Deployed to **AWS ECS Fargate** with **ALB routing**, **ECR registry**, and **Kubernetes manifests** for EKS deployment.

Job Portal REST API – Full-Stack Job Listing Platform **Oct 2025**
Python, FastAPI, SQLite, Docker, AWS ECS/ECR | GitHub

- Built **production-ready RESTful API** with **25+ endpoints** supporting full CRUD operations for job seekers, employers, postings, and applications with **Pydantic v2 validation**.
- Designed and implemented **HATEOAS-compliant architecture** enabling hypermedia-driven navigation, improving API discoverability and client integration.
- Containerized using **Docker Compose** with persistent volumes; deployed to **AWS ECR/ECS** demonstrating end-to-end **CI/CD pipeline**.

FieldFuze Backend – Enterprise Go API with Multi-Tenant RBAC **Dec 2024**
Go 1.23, Gin, DynamoDB, JWT, Docker, GitHub Actions

- Built **production-ready REST API** in **Go/Gin** with **multi-tenant RBAC architecture**, implementing complete user lifecycle (register, login, logout, validate).
- Designed **modular architecture** using **controller → service → repository pattern** with comprehensive middleware (JWT auth, CORS, logging).
- Created **infrastructure automation worker** for DynamoDB table/index management; wrote extensive **unit tests** for all layers.

Lambda Microservices Platform – Field Service Backend **Dec 2024**
Python, AWS Lambda, DynamoDB, S3, Stripe, Twilio, DocuSign, Terraform

- Developed **94 AWS Lambda functions** for complete SaaS platform, integrating **Stripe Connect** for vendor onboarding and payment processing.
- Implemented **DocuSign e-signature workflow**, **Twilio calling integration**, and **QuickBooks accounting sync** for invoicing automation.
- Built **automated deployment pipeline** with Git change detection using **Terraform IaC** and **CloudFormation** templates.

LuffyGPT – GPT Transformer Language Model from Scratch **Jan 2026 – Mar 2026**
Python, PyTorch, SentencePiece, tiktoken, Gradio, HuggingFace Hub | GitHub | Demo

- Implemented a **GPT decoder-only transformer from scratch** in PyTorch (10.81M params) — multi-head causal self-attention, positional embeddings, FFN with 4× expansion, residual connections, pre-LayerNorm, dropout; followed the *Attention is All You Need* architecture.
- Trained a **custom BPE tokenizer** using SentencePiece on a ~3.2M character corpus (vocab 2000, 3.11× compression); benchmarked char-level, GPT-2 BPE, cl100k, and o200k strategies; trained on **dual NVIDIA RTX 3090 GPUs** (DataParallel), reaching **val loss ~1.18** at 25K steps.
- Deployed as a **Gradio app on HuggingFace Spaces** with model weights hosted on HuggingFace Hub; built a CLI tool supporting `--train`, `--eval`, `--interactive`, and `--custom-bpe` modes.

EDUCATION

Lawrence Technological University **Jan 2024 – Dec 2025**
*Master of Science in Computer Science · GPA: **3.6/4.0*** *Southfield, MI*

- **Relevant Coursework: Machine Learning**, **Artificial Intelligence**, **Natural Language Processing**, **Intelligent Robotics (ROS)**, **Cloud Microservices**

Geethanjali College of Engineering & Technology **Aug 2018 – Aug 2022**
*Bachelor of Technology in Computer Science & Engineering · GPA: **7.5/10 (~3.0/4.0)*** *Hyderabad, Telangana*

- **Relevant Coursework: Software Engineering**, **Data Structures & Algorithms**, **Database Management**, **Operating Systems**

ACHIEVEMENTS

- **Selected for Amazon Nova AI Challenge: Trusted AI Track** (2025)
- **Built production SaaS platform** with 94 Lambda functions serving real customers (2024)
- **Gold Medalist in Indian National Mathematical Olympiad (INMO)** (2012)